When Nominal False-Positive Rates Fail: Key- and Length-Dependent Null Behavior of Canonical LLM Text Watermarks

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***Abstract*—Keyed text-watermark detectors are usually described by a nominal false-positive rate that follows from an idealized null: under the KGW green-list scheme, each scored token is green with probability γ independently of the others, so the detector's z-statistic is standard normal and $z > 2.326$ is a 1% test. We measure how far that null is from real text. Using exact reproductions of the maintained KGW SelfHash and SynthID-Text references, ten frozen detector keys, and 10,000 unwatermarked SmolLM2-135M outputs scored at 128, 256, and 512 tokens, we find that the nominal 1% KGW threshold has an empirical false-positive rate that depends on the key by two orders of magnitude: from 0.27% to 48.4% [47.4, 49.3] at 512 tokens, growing with length for the inflated keys. The mechanism is simple. Each key induces its own null green rate p_k on real text (0.205 to 0.302 against $\gamma = 0.25$), the resulting shift in mean z grows as $\sqrt{T}$ and predicts the observed shift with $r = 0.9999$, and over-dispersion adds the rest. The same keys scored on 5,000 human-written passages reproduce the key ranking (Spearman $\rho = +0.94$), so the bias is a property of the key against ordinary token statistics, not of the generating model, which only amplifies it. Which key inflates is model-specific: a Qwen2.5 replication fails different keys. Per-key, per-length empirical calibration on 5,000 outputs brings every held-out cell to 0.16% to 0.96%, and on watermarked text it costs a median two points of detection at 512 tokens while gaining detection for the keys whose threshold it lowers. A prospective familywise acceptance rule fitted at its own boundary nevertheless failed 18 of 60 cells, which we report as a design lesson on validation margin. SynthID-Text's naive-independence reference stays near 1% throughout. All rates are reported with exact 95% intervals.**

***Index Terms*—language-model watermarking, false-positive control, empirical calibration, KGW SelfHash, SynthID-Text, detector keys**

I. Introduction

A keyed text watermark lets a detector with the key test whether a text was produced by a compatible generation process. In the green-list family introduced by Kirchenbauer et al. [4], a secret key and the preceding context partition the vocabulary at each step into a green list of fraction γ and a red list; generation favours green tokens, and the detector counts how many of the T scored tokens are green. The detector's operating point is stated through a z-score whose null distribution is derived under one assumption: that in unwatermarked text each scored token is green with probability exactly γ, independently of every other. Under that assumption z is standard normal and $z > 2.326$ is a one-sided 1% test. Papers, toolkits, and deployment discussions routinely quote this nominal rate as if it were a property of the scheme.

This paper measures how far that null is from the behaviour of a real detector on real text, and explains the gap. We fix the implementation first: the canonical KGW SelfHash and SynthID-Text adapters used here reproduce the maintained reference implementations exactly, so nothing below can be attributed to a mismatch. We then score 10,000 unwatermarked SmolLM2-135M-Instruct outputs at three prefix lengths with ten frozen detector keys per scheme, report every rate with an exact 95% interval, and fit nothing.

That the i.i.d. assumption fails on real text is not new; Fernandez et al. [3] showed it and proposed corrected statistics. What has not been shown is how large the failure is, that its size and sign are set by the key, that it grows with length, that it survives on human-written text, and what correcting it costs in detection. Those are this paper's contributions, in order of evidential strength.

1) **The nominal 1% KGW threshold is key- and length-dependent.** Across ten keys at 512 tokens the empirical false-positive rate spans 0.27% to 48.4% [47.4, 49.3], $n = 10{,}000$ per cell; 20 of 30 key-by-length cells exceed 1% and the inflated keys get worse with length (Section IV-B). Which key inflates depends on the model: a Qwen2.5 replication fails different keys.
2) **The mechanism is a key-specific null green rate.** Each key colours a different fraction p_k of real text green, 0.205 to 0.302 against the assumed 0.25. Substituting p_k for γ in the detector's own formula predicts the mean null z of all 30 cells with $r = 0.9999$, and the shift grows as $\sqrt{T}$. Over-dispersion adds a second, smaller effect. Repetition is not the driver (Section IV-C).
3) **Human-written text reproduces the key ranking.** The same keys on 5,000 human-written passages give the same ordering of p_k (Spearman $\rho = +0.94$) and the same worst key, with smaller inflation. The bias

belongs to the key against ordinary token statistics; the model amplifies it (Section IV-D).

4) **Per-key, per-length empirical calibration works, and it is affordable.** Thresholds fitted once on 5,000 outputs and evaluated once on 5,000 disjoint outputs give held-out rates of 0.16% to 0.96% in all 60 scheme-by-key-by-length cells (Section IV-E). On watermarked text it costs a median two points of detection at 512 tokens, and for the three keys whose threshold it lowers it gains detection (Section IV-F).
5) **Fitting at an acceptance boundary leaves no familywise margin.** A prospective rule that required all 60 calibrated cells to pass an exact Bonferroni-controlled bound failed 18 cells, although no failed cell's point estimate exceeded 0.96%. Twenty-eight of 30 KGW thresholds had been fitted at the maximum passing count, where a fresh cell passes with probability about 55% (Section IV-G).

The contribution is deliberately bounded. We do not claim that every implementation called KGW or SynthID behaves alike, that the numbers transfer to other models or languages, or anything about robustness to editing or paraphrase. The detection rates we do report are development-scale and are there to price the remedy, not to claim a detection result. What we claim is narrower and, we think, more useful: for the named primitives, the nominal false-positive rate is not a property of the scheme, and the evaluation that replaces it must be indexed by key and by length.

II. Related Work

a) Keyed green-list watermarks.: Kirchenbauer et al. [4] introduced the green-list scheme and its z-test detector. The follow-up reliability study formalized context-dependent hashing, the SelfHash variant, repeated-context handling, and the dependence of detection on observed length [5]. Both derive the detector's null from the i.i.d. Bernoulli(γ) assumption examined here. Zhao et al. [14] fix the green list independently of context (Unigram) and prove robustness guarantees under the same style of null; Fernandez et al. [3] observed that the assumption fails on real text, proposed corrected statistics, and showed the resulting empirical false-positive inflation. Our contribution relative to that work is quantification: ten keys, three lengths, $n = 10{,}000$ per cell, exact intervals, an explicit per-key mechanism, and the human-text control.

b) Distortion-free and cryptographic constructions.: Kuditipudi et al. [6] and Christ et al. [1] construct watermarks whose output distribution matches the unwatermarked model and whose detectors have exact or provable null behaviour by design. SynthID-Text [2] uses context-dependent seeding and tournament sampling; its production evaluation covers quality, detection, edits, and paraphrasing. These families do not share KGW's z-test null, which is why our SynthID measurements are a reference rather than a failure and why the key effect is a KGW-specific finding.

c) Benchmarks and toolkits.: MarkMyWords [11], WaterBench [13], WaterPark [8], and WaterJudge [9] compare schemes on quality, detection, and tamper resistance; MarkLLM [10] packages many of them behind one interface. These evaluations typically report detection at a scheme's nominal threshold with one key. The present results show why that convention can misstate the operating point by an order of magnitude and why the key schedule belongs in the reported condition.

d) False-positive control.: Li et al. [7] develop pivotal statistics with explicit type-I error control for watermark detection, and forensic-readiness work argues that error rates must be established before watermark evidence is used in consequential settings [12]. We supply the empirical side: how large the error is for the most widely used detector, where it comes from, and what calibration removes it.

III. Methods

A. Implementation Identity

Before any null measurement, the validation suite pinned maintained reference revisions for KGW and SynthID-Text, checked keyed token assignments, reproduced aggregate detector scores from per-position traces, and compared generation step by step. The canonical adapters used throughout this paper are `kgw_author_selfhash_v1` (KGW SelfHash: $\gamma = 0.25$, $\delta = 2.0$, context width 4, repeated contexts ignored) and `synthid_deepmind_hash_v1`. The stock Transformers implementations were retained as separately named secondary variants and are never pooled with canonical scores. The remaining SynthID top-k boundary-tie difference was recorded as a decoder-policy difference.

B. Null Corpus and Detector Statistics

The primary null corpus is 10,000 unwatermarked continuations generated by the pinned SmolLM2-135M-Instruct checkpoint (temperature 0.8, top-k 40, 512 new tokens) from 10,000 distinct Databricks Dolly prompts balanced by task category. The corpus was produced as two disjoint 5,000-prompt splits for the calibration study of Section III-E; because nothing in the nominal-threshold analysis is fitted, both splits are pooled there. Every output was scored at its paired 128-, 256-, and 512-token prefixes by ten frozen detector keys per scheme. Keys within a run therefore score the same texts and are paired, not independent.

For KGW, the detector statistic at a prefix is

$$z = \frac{g - \gamma T}{\sqrt{T\gamma(1-\gamma)}}, \tag{1}$$

where g is the number of green tokens among the T eligible scored positions (SelfHash skips positions whose context has already been scored, so T is below the prefix length). The nominal one-sided 1% threshold is $z > 2.326$, which is exact only if green membership is i.i.d. Bernoulli(γ) under the null. SynthID-Text's mean-g detector has no nominal

z; as a reference we report the naive-independence normal approximation $z = (\bar{g} - 0.5)\sqrt{4\,d\,T}$ with d watermarking depth, at the same 2.326 cutoff, labelled as such.

All rates are reported with exact two-sided 95% Clopper–Pearson intervals. No threshold in Sections IV-B–IV-D was chosen from the data.

C. Secondary Null Runs

Two earlier runs with the same ten keys are reported alongside the primary corpus: a 500-output SmolLM2 development pilot, and an independent balanced 104-output run with Qwen2.5-0.5B-Instruct in a clean environment. Both are development diagnostics; their role here is to show that the primary result reproduces and that which key inflates is a property of the model.

D. Mechanism Analyses

Three detector-only analyses of the stored token ids ask why the nominal threshold fails. First, for each key k and length we pool green counts over eligible positions to estimate the null green rate p_k and compare the observed mean z with the value Eq. (1) implies when γ is replaced by p_k. Second, we relate each output's z to its repeated-4-gram fraction to test whether low-entropy repetition drives the inflation. Third, we score 5,000 de-duplicated human-written Dolly passages (`response` and `context` fields, at least 128 tokens, selected by a hashed seed) with the same ten keys at every prefix length each passage supports. No text is generated in any of these analyses.

E. Empirical Calibration and One-Shot Confirmation

The remedy we test is per-key, per-length empirical calibration. Thresholds were fitted once on the 5,000-output calibration split for 60 cells (two schemes, ten keys, three lengths) and then evaluated once on the disjoint 5,000-output confirmation split. Comparisons were strictly $s > t$. KGW thresholds were indexed by key and length; SynthID used one threshold per length equal to the maximum of the ten key-specific calibration candidates, with every key still tested separately at confirmation. The prospective acceptance rule was familywise: a cell passed if its exact one-sided Clopper–Pearson upper bound at $\alpha = 0.05/60$ was at most 1%, which allows at most 28 exceedances per cell, and the family passed only if all 60 cells passed. Thresholds were frozen and hashed before the confirmation split was generated, and no threshold was revised afterwards.

IV. Results

A. Canonical Parity Passed

The canonical adapters reproduced both pinned maintained references exactly on the tested detection fixtures and step-by-step generation checks, and aggregate scores reconstructed exactly from exported per-position rows. Every result below is therefore a property of the named primitives rather than of an implementation mismatch.

B. The Nominal 1% KGW Threshold Is Key- and Length-Conditional

Table I and Figure 1 give the empirical false-positive rate of canonical KGW SelfHash at $z > 2.326$ for all 30 key-by-length cells, $n = 10{,}000$ each. Point estimates exceed 1% in 20 of 30 cells; the exact 95% interval lies entirely above 1% in 19 cells and entirely below 1% in 8. The spread across keys is two orders of magnitude at every length. Key 07 gives 19.1%, 33.0%, and 48.4% [47.4, 49.3] at 128, 256, and 512 tokens; key 03 gives 0.32%, 0.26%, and 0.27%. The median across keys rises from 3.5% at 128 tokens to 6.9% at 512. Inflated keys therefore get worse with length while conservative keys stay conservative, so no single length policy repairs the threshold. The 500-output pilot value of 47.0% for key 07 at 512 tokens reproduces as 48.4% at $n = 10{,}000$.

Which key inflates is model-specific (Figure 2). On Qwen2.5-0.5B the worst keys are 04 and 05 at 41.3% each, while key 07, the worst key on SmolLM2, gives 5.8%. The ten keys are the same bit strings in both runs; the model's output distribution decides which of them fail.

SynthID-Text's naive-independence reference behaves differently: per-cell rates range from 0.74% to 1.58%, 11 of 30 cells sit above 1% by point estimate, and seven have intervals entirely above 1%. This is mild over-dispersion with no key-specific bias, and we report it as a reference rather than as a SynthID failure.

C. Mechanism: Each Key Has Its Own Null Green Rate

The per-key null z distributions at 512 tokens (Figure 3) are approximately normal but shifted and widened relative to the assumed $N(0,1)$: mean z runs from -1.96 (key 03) to $+2.24$ (key 07), and the standard deviation from 1.26 to 1.58.

The shift has a simple source (Figure 4). Each key induces its own null green-token rate p_k on this model's text, stable across lengths and ranging from 0.205 to 0.302 at 512 tokens against the assumed $\gamma = 0.25$. Substituting p_k for γ in Eq. (1) predicts a mean null z of $(p_k - \gamma)\sqrt{T}/\sqrt{\gamma(1-\gamma)}$, which grows as $\sqrt{T}$. The observed mean z across all 30 cells matches this prediction with $r = 0.9999$. A second, smaller effect is over-dispersion: the standard deviation of z rises from 1.07–1.27 at 128 tokens to 1.26–1.58 at 512, which is why key 06 ($p_k = 0.249$) still shows 3.7% at 512 tokens. A normal with each cell's observed mean and standard deviation reproduces every cell's rate within 1.5 percentage points.

The failure is thus the standard z-test's i.i.d. Bernoulli(γ) null being wrong for real text, with the sign and size of the error set by which frequent (context, token) pairs each key colours green. This is the phenomenon Fernandez et al. [3] documented for KGW-style detectors; here it is quantified per key and per length on one pinned implementation.

Repetition is not the driver (Figure 5). The outputs are repetitive (median repeated-4-gram fraction 0.23), but for

TABLE I
EMPIRICAL FALSE-POSITIVE RATE (%) OF CANONICAL KGW SELFHASH AT THE NOMINAL ONE-SIDED 1% THRESHOLD ($z > 2.326$) ON 10,000 UNWATERMARKED SMOLLM2-135M-INSTRUCT OUTPUTS PER CELL, WITH EXACT TWO-SIDED 95% CLOPPER–PEARSON INTERVALS. NOTHING IS FITTED.

Key	128 tokens	256 tokens	512 tokens
kgw-00	6.23 [5.76, 6.72]	8.87 [8.32, 9.44]	12.49 [11.85, 13.15]
kgw-01	0.75 [0.59, 0.94]	0.70 [0.55, 0.88]	0.80 [0.63, 0.99]
kgw-02	0.99 [0.81, 1.20]	1.19 [0.99, 1.42]	1.28 [1.07, 1.52]
kgw-03	0.32 [0.22, 0.45]	0.26 [0.17, 0.38]	0.27 [0.18, 0.39]
kgw-04	0.66 [0.51, 0.84]	0.80 [0.63, 0.99]	0.88 [0.71, 1.08]
kgw-05	4.80 [4.39, 5.24]	7.17 [6.67, 7.69]	10.19 [9.60, 10.80]
kgw-06	2.18 [1.90, 2.49]	2.78 [2.47, 3.12]	3.69 [3.33, 4.08]
kgw-07	19.08 [18.31, 19.86]	33.02 [32.10, 33.95]	48.36 [47.38, 49.34]
kgw-08	7.28 [6.78, 7.81]	9.65 [9.08, 10.25]	12.74 [12.09, 13.41]
kgw-09	6.44 [5.97, 6.94]	9.30 [8.74, 9.89]	13.66 [12.99, 14.35]
Median	3.49	4.98	6.94

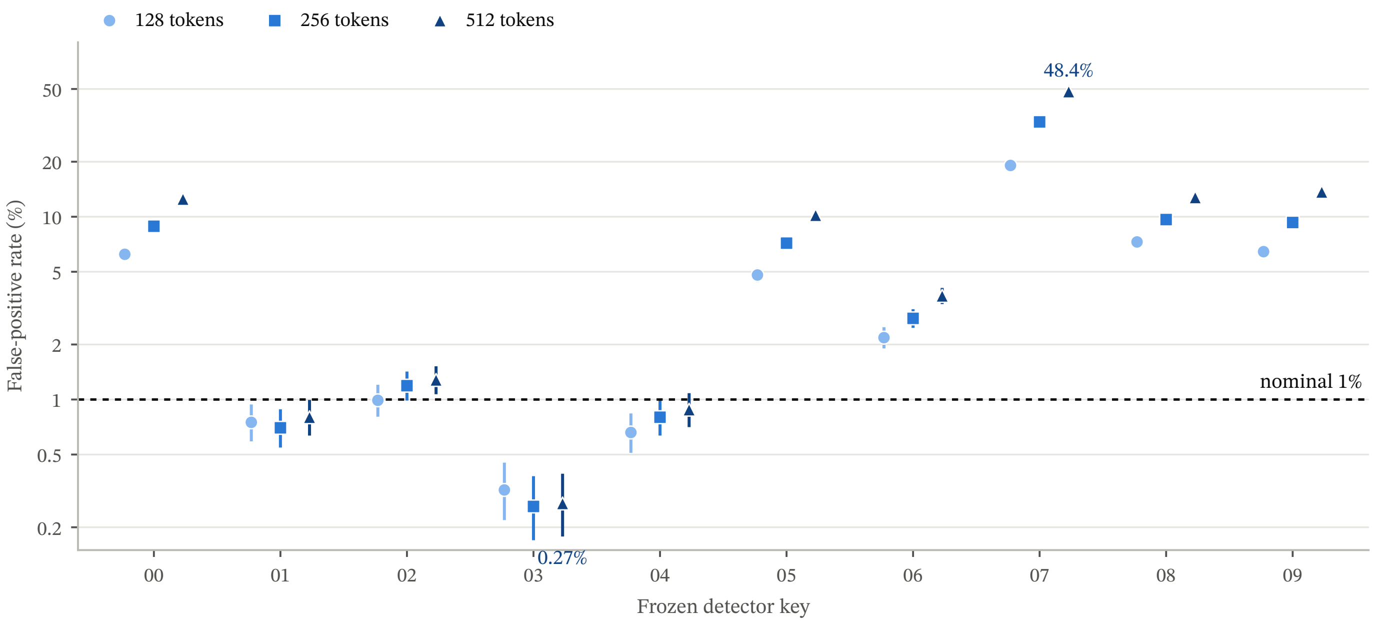


Fig. 1. Canonical KGW SelfHash false-positive rate by frozen detector key and prefix length at the nominal $z > 2.326$ threshold, $n = 10{,}000$ unwatermarked SmolLM2 outputs per cell, exact 95% intervals, log scale. Prefix length is shaded light to dark. Keys are categorical and are paired within texts, so the points are not independent across keys.

the worst key the correlation between z and repetition is negative (Spearman $\rho = -0.29$): key 07 gives 63% in the least repetitive quartile and 32% in the most repetitive. SelfHash already skips repeated contexts, which is why eligible positions average 363 of 512.

D. Human-Written Text Reproduces the Key Ranking

If p_k were a property of the generating model, human-written text under the same keys would show a different ranking. It does not (Table II, Figure 6). On 5,000 human-written Dolly passages, p_k ranges from 0.216 to 0.280 at 512 tokens and tracks the model's p_k key for key (Spearman $\rho = +0.94$, $p = 5 \times 10^{-5}$; Pearson $r = +0.96$ over ten keys). Key 07 is the worst key in both populations and keys 03 and 04 the best in both. Human-text rates at the nominal threshold are inflated in the same direction but by less: key 07 gives 9.4%, 16.4%, and 25.5% at 128, 256, and 512 tokens on human text against 19.1%, 33.0%, and 48.4% on model output, and 19 of 30 human cells lie above 1% by point estimate. Human text is also far less repetitive (median repeated-4-gram fraction 0.016), and the inflation persists.

The key-specific bias is therefore a property of each key's green list against ordinary English token statistics, not of SmolLM2's output distribution. The model amplifies it, presumably because its low-entropy output concentrates mass on the same frequent (context, token) pairs. A practical corollary is that a deployment cannot repair the nominal

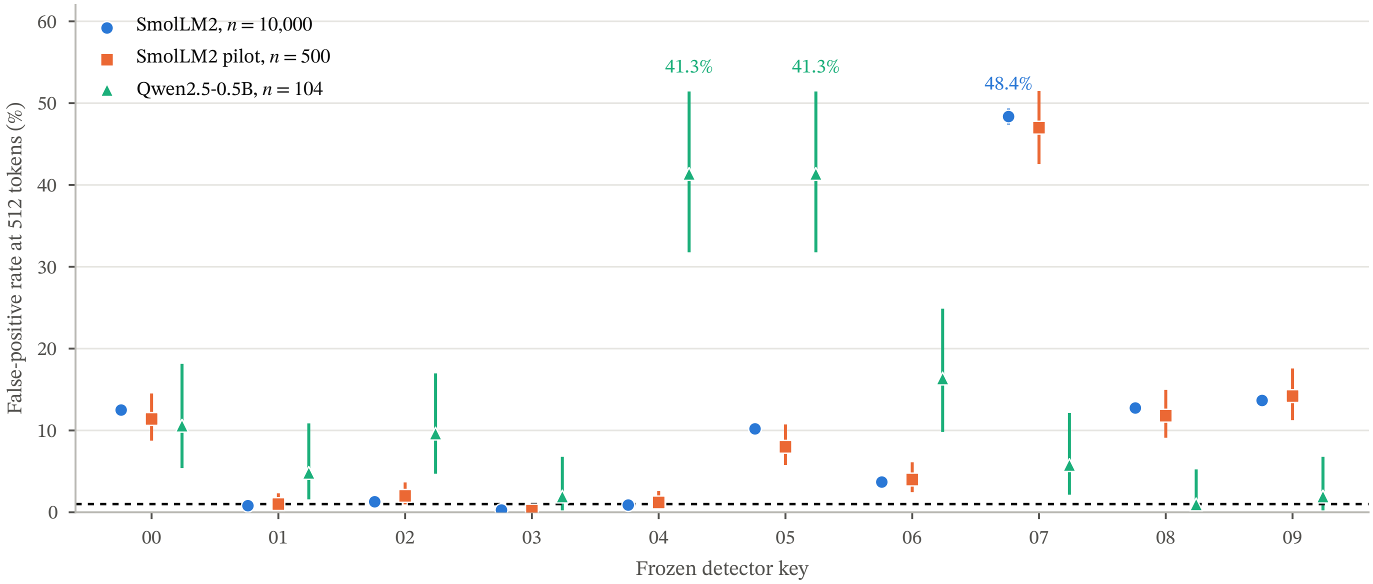


Fig. 2. The same ten keys at 512 tokens in three null runs: the primary SmolLM2 corpus ($n = 10{,}000$), the SmolLM2 pilot ($n = 500$), and the Qwen2.5-0.5B replication ($n = 104$). The two SmolLM2 runs agree; the Qwen run inflates different keys.

TABLE II
EMPIRICAL FALSE-POSITIVE RATE (%) AT THE NOMINAL $z > 2.326$ THRESHOLD ON HUMAN-WRITTEN DOLLY PASSAGES (HUMAN) AND ON UNWATERMARKED SMOLLM2 OUTPUT (MODEL) FOR THE SAME TEN FROZEN KGW KEYS. EACH HUMAN PASSAGE IS SCORED AT EVERY PREFIX LENGTH IT SUPPORTS, SO n FALLS WITH LENGTH. THE LAST ROWS GIVE THE RANGE OF THE NULL GREEN RATE p_k ACROSS KEYS AND THE CELL SIZES.

	128 tokens		256 tokens		512 tokens	
Key	Human	Model	Human	Model	Human	Model
kgw-00	2.6	6.2	2.7	8.9	2.7	12.5
kgw-01	0.7	0.8	0.5	0.7	0.5	0.8
kgw-02	0.9	1.0	0.9	1.2	1.4	1.3
kgw-03	0.5	0.3	0.9	0.3	1.3	0.3
kgw-04	0.2	0.7	0.2	0.8	0.7	0.9
kgw-05	3.6	4.8	5.4	7.2	7.2	10.2
kgw-06	1.4	2.2	1.0	2.8	1.8	3.7
kgw-07	9.4	19.1	16.4	33.0	25.5	48.4
kgw-08	6.2	7.3	9.2	9.7	14.6	12.7
kgw-09	2.9	6.4	4.8	9.3	6.9	13.7
p_k range	0.217–0.282	0.206–0.301	0.217–0.282	0.206–0.302	0.216–0.280	0.205–0.302
n	5,000	10,000	1,951	10,000	554	10,000

threshold by calibrating on model text alone: human text under the same key inflates too.

E. Per-Key, Per-Length Calibration Brings Held-Out Rates Below 1%

Appendix Table III and Table IV list all 60 calibrated cells. Fitted KGW thresholds at 512 tokens range from $z = 1.49$ (key 03) to $z = 6.40$ (key 07) against the nominal 2.326. On the disjoint confirmation split the held-out false-positive rate of the once-fitted thresholds lies between 0.16% and 0.96% in every cell, with no cell above 1% by point estimate; the worst cell is key 03 at 512 tokens, 0.96% [0.71, 1.27]. KGW cells range 0.24% to 0.96% and SynthID cells 0.16% to 0.66%. Empirical calibration at $n = 5{,}000$ per cell thus delivers the operating point the nominal threshold promised, provided the threshold is indexed by key and length.

F. What Calibration Costs in Detection

A per-key threshold that is far above the nominal one invites an obvious objection: for key 07 at 512 tokens it sits at $z > 6.40$ against a nominal 2.326, and a threshold that high might detect nothing. We can answer this from stored scores. A 1,000-output development screen generated watermarked text under the same model, the same watermark configuration and the same ten-key schedule,

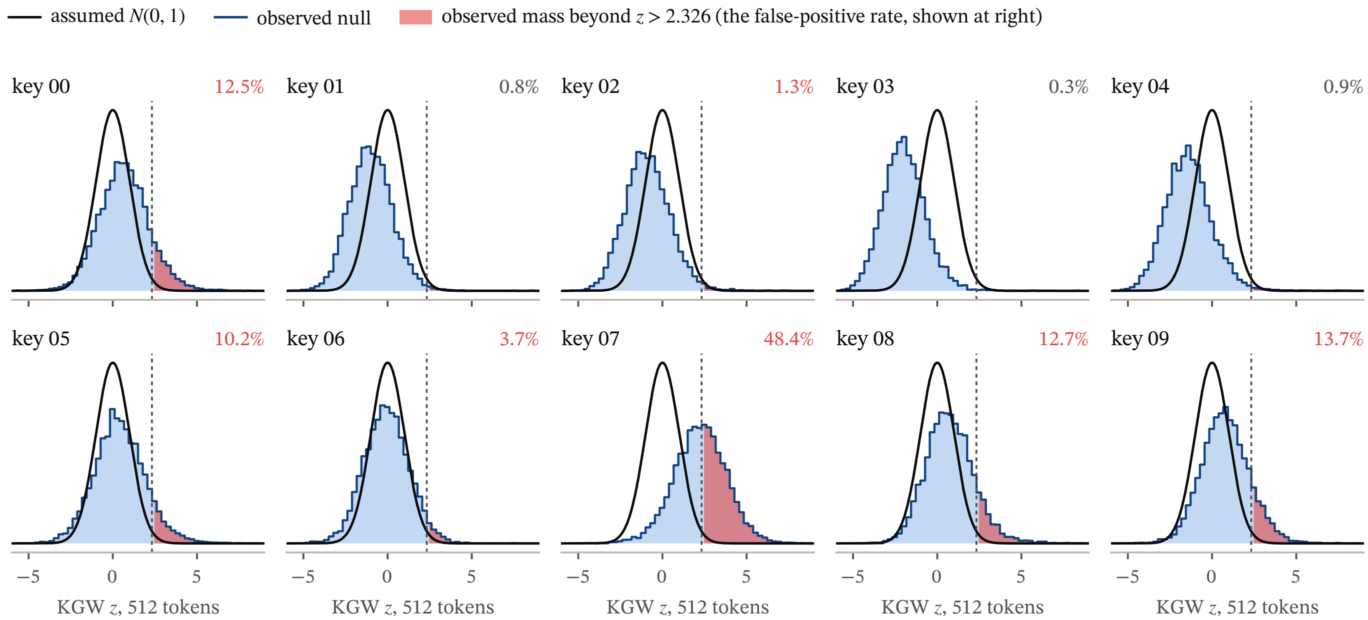


Fig. 3. Null distribution of the KGW z statistic at 512 tokens for each of the ten keys, n = 10,000, against the assumed standard normal (solid) and the nominal cutoff (dashed). The shaded area past the cutoff is the mass this key puts beyond the nominal 1% threshold; the percentage at each panel's top right is that cell's false-positive rate. Under the assumed null every panel would show the same 1%.

so the thresholds frozen on the null's calibration split apply to it unchanged. Those thresholds were fitted on unwatermarked text alone, so using them on watermarked text is not circular. Detection rate here is the fraction of watermarked outputs scoring above the threshold under their own key, on 50 outputs per cell.

Calibration does not disarm the detector (Figure 7a). The median detection rate across the ten keys falls from 99% to 97% at 512 tokens, from 98% to 95% at 256, and from 96% to 88% at 128. For key 07 at 512 tokens the false-positive rate falls from 48.4% to 0.54% while detection falls from 98% to 96%: an almost ninetyfold reduction in false positives for two points of detection. The worst cell in the whole set is key 08 at 128 tokens, where detection is 76% [62, 87].

The correction also runs in both directions (Figure 7b). Calibration raised the threshold in 27 of 30 cells and lowered it in three, and where it lowered the threshold, detection improved. Key 03 is the clearest case: its null sits well below the assumed one, its calibrated threshold at 512 tokens is $z > 1.49$ rather than 2.326, and its detection rate rises from 96% to 98%; at 128 tokens it rises from 78% to 86%. Averaged over cells, calibration costs 4.2 points of detection where it raises the threshold and gains 5.3 points where it lowers it. A conservative key is not harmless. It gives away detection power for a false-positive rate below the one it was asked for, and only a key-conditional threshold recovers that.

These are development-scale numbers on 50 outputs per cell, so the intervals are wide, and they are not a confirmatory detection claim. They are sufficient to answer the objection they were computed for: on this model and this watermark configuration, the price of a correctly calibrated false-positive rate is a few points of detection, not the loss of the watermark.

G. Fitting at the Acceptance Boundary Leaves No Familywise Margin

The same 60 cells were also subject to the prospective familywise rule of Section III-E, and that rule failed: 42 cells passed and 18 did not (14 KGW, 4 SynthID), even though every failed cell's point estimate lay between 0.58% and 0.96% and the largest Bonferroni-controlled upper bound was 1.48%. The reason is a design flaw rather than a held-out shift. Twenty-eight of 30 KGW calibration cells were fitted at exactly the maximum passing count of 28 exceedances, and KGW averaged 27.9 exceedances in calibration against 27.7 in confirmation (Figure 8a). At a true rate equal to the boundary value $28/5{,}000 = 0.56\%$, a fresh 5,000-output cell has only about a 55% probability of staying at or below the passing count, so a rule that requires all 60 cells to pass was designed to fail. Prospective exact-binomial planning shows that more observations narrow the penalty but do not remove the need for a true operating rate below the target: under a conservative union bound, the largest true per-cell rate compatible with 95% whole-family pass probability rises from about 0.30% at 5,000 observations to 0.74% at 50,000 (Figure 8b). These are planning scenarios, not replacement thresholds.

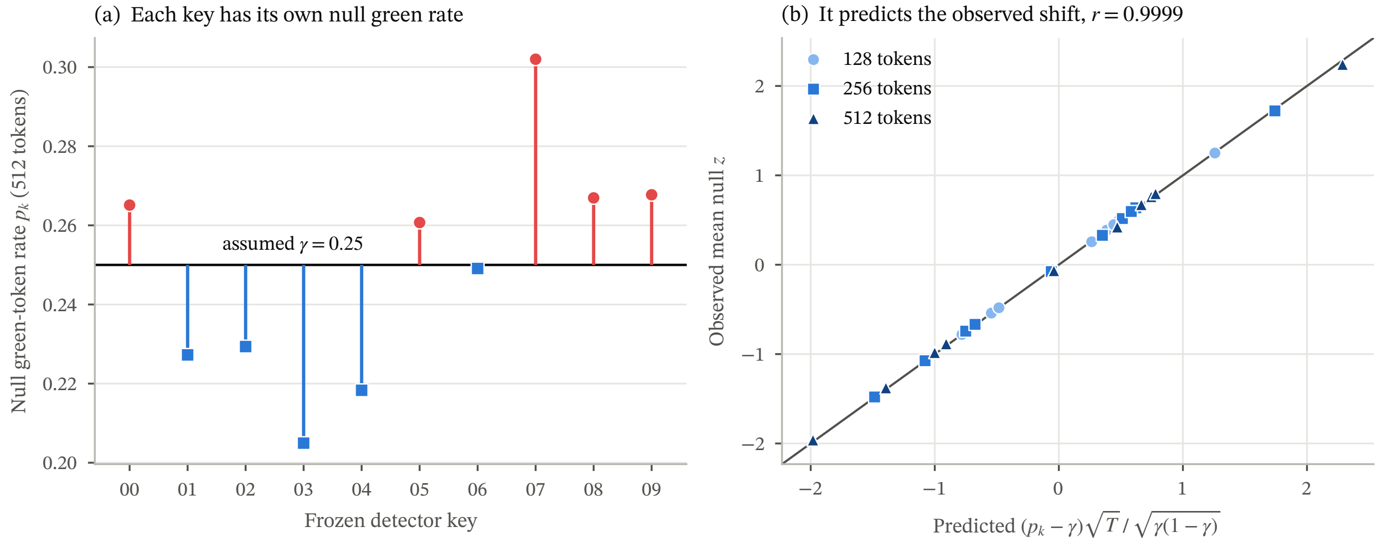


Fig. 4. (a) Null green-token rate p_k for each key at 512 tokens, drawn from the assumed $\gamma = 0.25$; colour marks which side of γ the key falls on and the vertical bars are exact 95% intervals. (b) Observed mean null z for all 30 key-by-length cells against the value predicted from p_k alone, with the identity line.

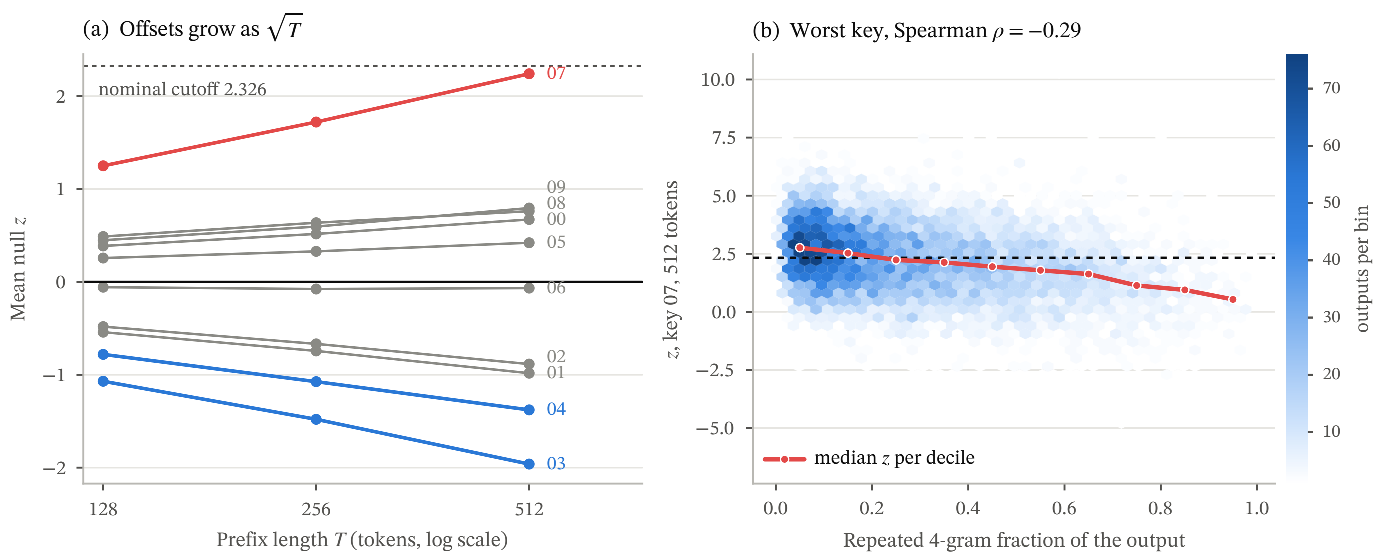


Fig. 5. (a) Mean null z per key against prefix length on a log scale; the offsets grow as $\sqrt{T}$, away from zero in both directions, and only the keys ending beyond $|z| = 1$ are coloured. (b) Density of per-output z for key 07 at 512 tokens against the output's repeated-4-gram fraction, $n = 10{,}000$, with the median per decile; more repetition goes with lower z, not higher.

V. Discussion

A. The Nominal Rate Is Not a Property of the Scheme

The i.i.d. Bernoulli(γ) null treats the green list as a random partition of the vocabulary that ordinary text does not prefer. It is not. Each key's partition lands differently on the frequent (context, token) pairs of English, so each key has its own p_k, and the z-test amplifies the difference by $\sqrt{T}$. Two consequences follow. A scheme-level false-positive rate is meaningless without the key: for the same model and threshold, one key gives 0.27% and another 48.4%. And longer texts do not help: for an inflated key the nominal test gets worse with length, because the bias grows with T while the assumed standard deviation does not.

B. The Bias Belongs to the Key, the Amplification to the Model

Human-written text under the same keys reproduces the ranking of p_k and the worst key, with smaller deviations from γ. This separates two explanations that the model-only data could not: the key effect is not an artefact of a tiny, repetitive model, and the model's contribution is to concentrate its output on the pairs the key already favours.

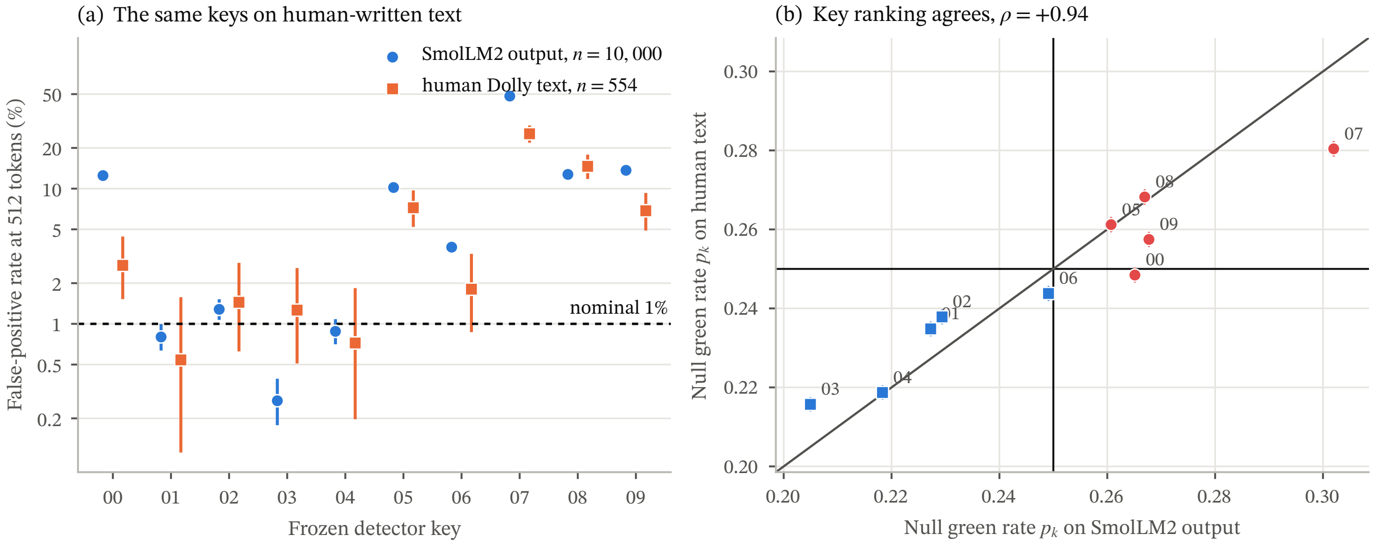


Fig. 6. (a) False-positive rate at the nominal threshold and 512 tokens for the same ten keys on SmolLM2 output (n = 10,000) and on human-written Dolly passages (n = 554), exact 95% intervals, log scale. The human intervals are wide because the 512-token human cell is small. (b) Per-key null green rate on human text against the same quantity on model output, with exact intervals on both axes; the diagonal is equality and the rules mark $\gamma = 0.25$. Colour is the same side-of-γ encoding as Figure 4a.

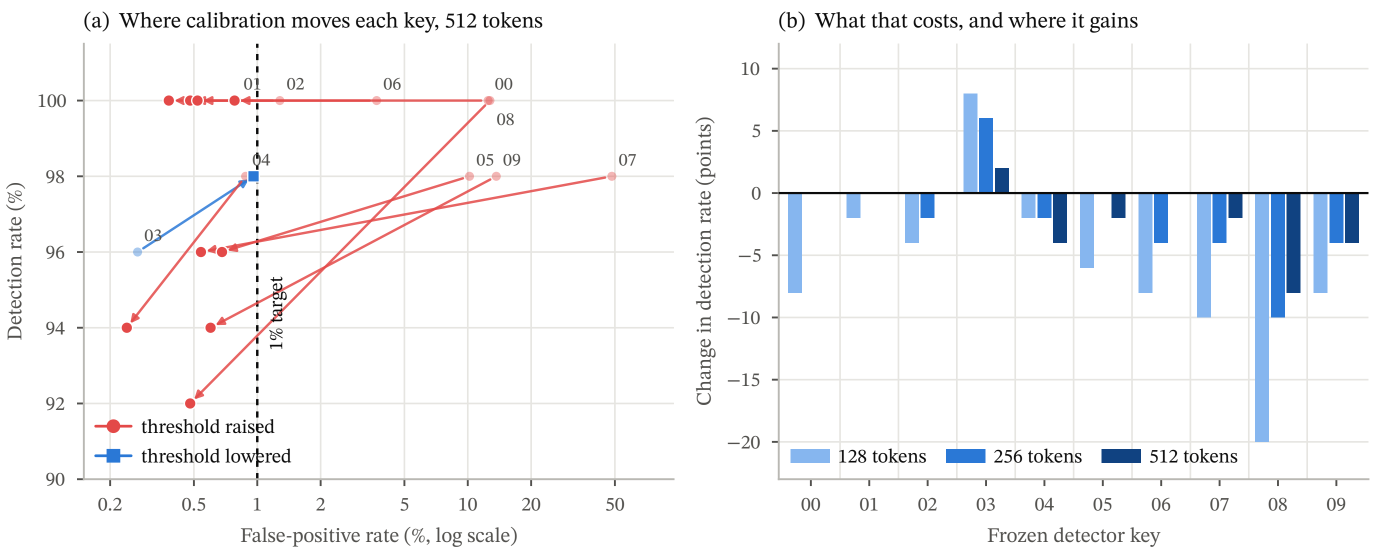
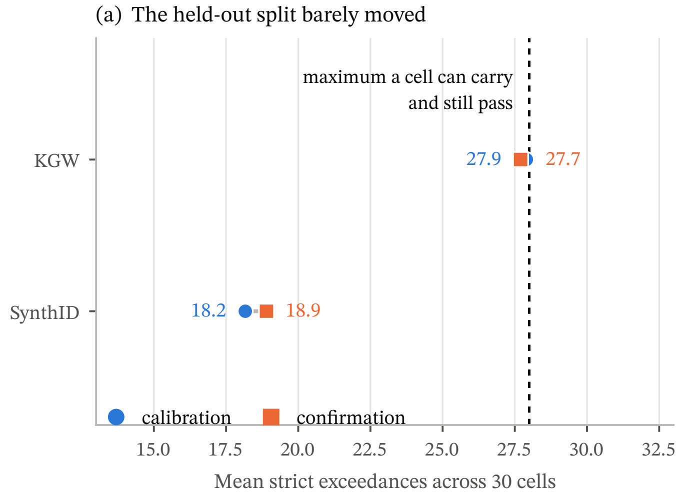


Fig. 7. (a) Operating point of each detector key at 512 tokens before and after per-key calibration, on a log false-positive axis. Arrows run from the nominal threshold to the calibrated one; labels sit at the nominal end. Colour marks whether calibration raised or lowered that key's threshold. (b) Change in detection rate from calibration, by key and prefix length. The three cells where the threshold fell are the three where detection improved. Detection rates are development-scale, 50 watermarked outputs per cell.

The Qwen2.5 replication shows the other side of the same mechanism: a different model has a different frequent-pair distribution, and different keys fail. Both facts argue for the same practice. A deployment should measure p_k, or equivalently the null rate, for its own key on the text population it will score, including human text, rather than inherit a rate from the scheme.

C. Calibration Works; Familywise Guarantees Need Margin

Per-key, per-length thresholds fitted once on 5,000 outputs delivered held-out rates below 1% in every cell. That is the practical remedy, and it needs nothing beyond an unwatermarked corpus from the relevant population and a key schedule. It is also cheap in the currency practitioners care about: a few points of detection where the threshold rises, and more detection than the nominal threshold gave where it falls. The prospective familywise rule we also imposed failed, and the failure is instructive. Selecting the least conservative threshold that satisfies a calibration bound places every cell at the acceptance boundary, where a fresh sample passes with probability near one half; requiring all 60 to pass then has little chance of success however valid



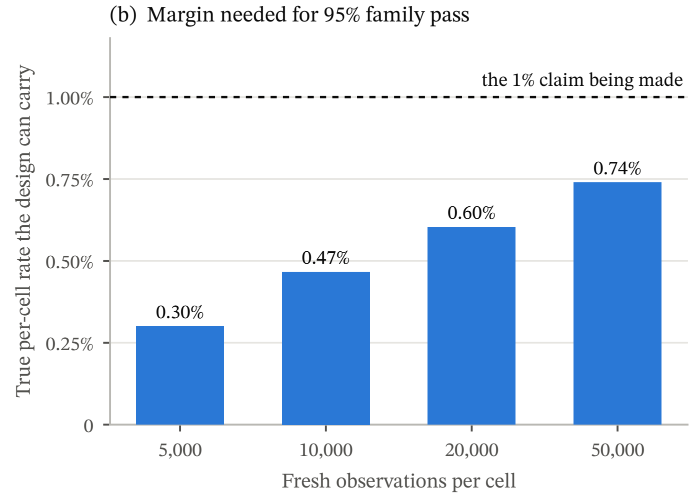


Fig. 8. (a) Mean strict exceedances across the 30 key-by-length cells of each scheme, on the calibration split and on the disjoint confirmation split, against the largest count a cell could carry and still pass. KGW's pair sits on the boundary and barely moves, so the gate failure is not a held-out shift. (b) Prospective union-bound planning: the largest true per-cell rate compatible with at least 95% probability that all 60 cells pass, by fresh sample size. These are design scenarios, not replacement thresholds.

the multiplicity control. Protocols that want a simultaneous guarantee should declare both the final error target and a lower design rate that gives the family adequate pass probability, and should size the calibration corpus to that design rate rather than to the target.

D. Implications for Operational Use

A keyed detector score is evidence of compatibility with a specified embedding and detection process. It is not proof of authorship, misconduct, truth, or intent. Operational claims should name the scheme variant, the key policy, the length range, the null population on which the threshold was calibrated, the uncertainty procedure, and the consequences of a false positive. On the evidence here, a claim that rests on the nominal rate alone is unsupported for KGW-style detectors.

VI. Limitations

The primary experiments use small language models and pinned English prompt sources; the size of the effect will differ for other models, languages, and domains, although the mechanism does not depend on model size. Findings apply to the explicitly named variants and parameter settings rather than to every implementation called KGW or SynthID. The Qwen2.5 replication is small ($n = 104$) and establishes only that the failing keys differ. The human-text null has 554 passages at 512 tokens, and its longer passages are largely reference text rather than answers. SynthID-Text was evaluated against a naive-independence reference because its mean-g detector has no nominal z; its calibrated behaviour is in Appendix A. The detection rates in Section IV-F come from a 50-output-per-cell development screen, so they price the remedy at development scale rather than confirming a detection claim. No robustness, attack or paraphrase conclusion is available.

VII. Ethics and Responsible Use

False accusations based on automated text detection can cause material harm. Accordingly, all claims in this paper remain conditional on named implementations, keys, lengths, populations, and decision rules. We discourage use of watermark detection as standalone evidence in disciplinary, employment, legal, or authorship decisions. Raw generated continuations will be reviewed for memorized personal, copyrighted, or unsafe content before any public release.

VIII. Reproducibility and Artifact Availability

The repository records frozen configurations, model and dataset revisions, prompt selection manifests, key schedules, implementation fingerprints, per-position audit formats, the frozen calibration thresholds, and the confirmation record. Every table and figure in this paper is generated deterministically from stored scores by a fixed script; no analysis required new generation. Model weights and cached datasets are not redistributed; deterministic retrieval instructions and immutable revisions are supplied instead.

The release is at https://github.com/nisaharan/ai-watermarks-3d and includes source code, tests, aggregate artifacts, prompt row identifiers and hashes, tables, figures, and parity fixtures. It also carries the two compressed score files the analyses read, so a clone regenerates every table and figure in this paper without generating any text; this was checked by cloning the release and rebuilding the paper from it alone. The text-level tables carry token statistics and a SHA-256 per text, never the text itself. Raw Dolly text requires CC BY-SA 3.0 attribution and share-alike treatment. Raw generated continuations and development runs that support no claim in this paper are retained privately for audit and excluded from the reported package.

IX. Conclusion

For canonical KGW SelfHash, the nominal 1% false-positive threshold is neither 1% nor a constant: it is a key- and length-dependent quantity that ranged from 0.27% to 48.4% on 10,000 unwatermarked outputs, because each key induces its own null green rate on real text and the detector's z-test scales the resulting bias by $\sqrt{T}$. Human-written text reproduces the key ranking, so the effect is a property of the key rather than the model. Empirical calibration indexed by key and length removes it. Evaluations and deployments of green-list watermarks should therefore report the key schedule and the calibration population as part of the experimental condition, and should treat a nominal rate as a hypothesis to be tested rather than a property to be quoted.

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Appendix A
Calibrated Per-Cell Held-Out False-Positive Rates

Table III lists the 30 canonical KGW cells and Table IV the 30 SynthID-Text cells. Each row gives the threshold fitted on the calibration split, the strict exceedance count on both splits, and the held-out rate on the confirmation split with its exact interval. No cell's held-out point estimate exceeds 1%.

Appendix B
Data Sources and Evidential Role

Table V lists every corpus the paper uses and what each one is for. All scoring is detector-only on stored token ids.

Appendix C
Interpretation Checklist

- Every rate is conditional on the named variant, key, prefix length, threshold, and null population; none transfers to another key by default.
- A cell that failed the familywise rule in Section IV-G is not evidence that its true rate exceeds 1%; its point estimate is in Appendix A.
- The SynthID-Text figures are a naive-independence reference, not a nominal claim by that scheme.
- Canonical and Transformers variant scores are never pooled.
- The detection rates are a development screen of 50 outputs per cell, reported to price calibration; they are not a confirmatory detection claim.
- Robustness to editing, paraphrase or attack was not evaluated.

TABLE III

Canonical KGW SelfHash (thresholds indexed by key and length). Held-out false-positive rate of the once-fitted threshold on the 5,000-output confirmation split, with exact two-sided 95% Clopper–Pearson intervals. Calib. and Conf. are strict exceedance counts on the calibration and confirmation splits; the Bonferroni-controlled one-sided exact upper bound had to be at most 1% (28 exceedances) for the cell to pass.

Key	T	Threshold	Calib.	Conf.	Held-out FPR (%) [95% CI]	Upper bound (%)	Pass
kgw-00	128	3.8636	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-00	256	4.3585	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-00	512	5.0208	28	26	0.52 [0.34, 0.76]	0.92	yes
kgw-01	128	2.4271	27	28	0.56 [0.37, 0.81]	0.98	yes
kgw-01	256	2.6469	28	17	0.34 [0.20, 0.54]	0.69	yes
kgw-01	512	2.7107	28	19	0.38 [0.23, 0.59]	0.74	yes
kgw-02	128	2.7406	28	14	0.28 [0.15, 0.47]	0.60	yes
kgw-02	256	3.0645	28	17	0.34 [0.20, 0.54]	0.69	yes
kgw-02	512	3.0632	28	24	0.48 [0.31, 0.71]	0.87	yes
kgw-03	128	1.9261	26	32	0.64 [0.44, 0.90]	1.08	no
kgw-03	256	1.7457	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-03	512	1.4927	28	48	0.96 [0.71, 1.27]	1.48	no
kgw-04	128	2.4444	28	24	0.48 [0.31, 0.71]	0.87	yes
kgw-04	256	2.5404	28	35	0.70 [0.49, 0.97]	1.15	no
kgw-04	512	3.2515	28	12	0.24 [0.12, 0.42]	0.55	yes
kgw-05	128	3.6977	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-05	256	4.1435	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-05	512	4.8658	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-06	128	2.9673	28	32	0.64 [0.44, 0.90]	1.08	no
kgw-06	256	3.0995	28	37	0.74 [0.52, 1.02]	1.20	no
kgw-06	512	3.3158	28	39	0.78 [0.56, 1.06]	1.25	no
kgw-07	128	4.8488	28	23	0.46 [0.29, 0.69]	0.85	yes
kgw-07	256	5.6614	28	16	0.32 [0.18, 0.52]	0.66	yes
kgw-07	512	6.3967	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-08	128	4.3146	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-08	256	4.9333	28	30	0.60 [0.41, 0.86]	1.03	no
kgw-08	512	5.6159	28	24	0.48 [0.31, 0.71]	0.87	yes
kgw-09	128	3.7778	28	29	0.58 [0.39, 0.83]	1.00	no
kgw-09	256	4.2699	28	31	0.62 [0.42, 0.88]	1.05	no
kgw-09	512	4.7962	28	30	0.60 [0.41, 0.86]	1.03	no

TABLE IV
CANONICAL SYNTHID-TEXT (ONE SHARED THRESHOLD PER LENGTH). HELD-OUT FALSE-POSITIVE RATE OF THE ONCE-FITTED THRESHOLD ON THE 5,000-OUTPUT CONFIRMATION SPLIT, WITH EXACT TWO-SIDED 95% CLOPPER–PEARSON INTERVALS. CALIB. AND CONF. ARE STRICT EXCEEDANCE COUNTS ON THE CALIBRATION AND CONFIRMATION SPLITS; THE BONFERRONI-CONTROLLED ONE-SIDED EXACT UPPER BOUND HAD TO BE AT MOST 1% (28 EXCEEDANCES) FOR THE CELL TO PASS.

Key	T	Threshold	Calib.	Conf.	Held-out FPR (%) [95% CI]	Upper bound (%)	Pass
synthid-00	128	0.5456	12	12	0.24 [0.12, 0.42]	0.55	yes
synthid-00	256	0.5344	27	21	0.42 [0.26, 0.64]	0.79	yes
synthid-00	512	0.5293	23	16	0.32 [0.18, 0.52]	0.66	yes
synthid-01	128	0.5456	10	14	0.28 [0.15, 0.47]	0.60	yes
synthid-01	256	0.5344	15	20	0.40 [0.24, 0.62]	0.77	yes
synthid-01	512	0.5293	16	21	0.42 [0.26, 0.64]	0.79	yes
synthid-02	128	0.5456	9	15	0.30 [0.17, 0.49]	0.63	yes
synthid-02	256	0.5344	16	14	0.28 [0.15, 0.47]	0.60	yes
synthid-02	512	0.5293	24	16	0.32 [0.18, 0.52]	0.66	yes
synthid-03	128	0.5456	28	30	0.60 [0.41, 0.86]	1.03	no
synthid-03	256	0.5344	20	32	0.64 [0.44, 0.90]	1.08	no
synthid-03	512	0.5293	28	31	0.62 [0.42, 0.88]	1.05	no
synthid-04	128	0.5456	19	17	0.34 [0.20, 0.54]	0.69	yes
synthid-04	256	0.5344	28	22	0.44 [0.28, 0.67]	0.82	yes
synthid-04	512	0.5293	27	25	0.50 [0.32, 0.74]	0.90	yes
synthid-05	128	0.5456	15	9	0.18 [0.08, 0.34]	0.46	yes
synthid-05	256	0.5344	14	20	0.40 [0.24, 0.62]	0.77	yes
synthid-05	512	0.5293	17	22	0.44 [0.28, 0.67]	0.82	yes
synthid-06	128	0.5456	6	8	0.16 [0.07, 0.32]	0.43	yes
synthid-06	256	0.5344	14	14	0.28 [0.15, 0.47]	0.60	yes
synthid-06	512	0.5293	19	16	0.32 [0.18, 0.52]	0.66	yes
synthid-07	128	0.5456	17	12	0.24 [0.12, 0.42]	0.55	yes
synthid-07	256	0.5344	17	17	0.34 [0.20, 0.54]	0.69	yes
synthid-07	512	0.5293	23	21	0.42 [0.26, 0.64]	0.79	yes
synthid-08	128	0.5456	11	13	0.26 [0.14, 0.44]	0.57	yes
synthid-08	256	0.5344	27	25	0.50 [0.32, 0.74]	0.90	yes
synthid-08	512	0.5293	18	33	0.66 [0.45, 0.93]	1.10	no
synthid-09	128	0.5456	13	12	0.24 [0.12, 0.42]	0.55	yes
synthid-09	256	0.5344	16	19	0.38 [0.23, 0.59]	0.74	yes
synthid-09	512	0.5293	16	20	0.40 [0.24, 0.62]	0.77	yes

TABLE V
EVERY CORPUS USED IN THIS PAPER AND THE ROLE IT PLAYS. ALL SCORING IS DETECTOR-ONLY ON STORED TOKEN IDS.

Corpus	Size	Role
SmolLM2-135M unwatermarked outputs	10,000	Primary null (Sections IV-B–IV-C); also the calibration and confirmation splits of 5,000 each (Sections IV-E–IV-G).
SmolLM2 development pilot	500	Secondary run; reproduces the primary key profile.
Qwen2.5-0.5B replication	104	Secondary run; shows that the failing keys are model-specific.
Human-written Dolly passages	5,000	Human-text null (Section IV-D); detector-only, no generation.
Development detection-rate screens	2,400	Out of scope; support no claim here.